Analysis of the MOS Transistor Based on the Self-Consistent Solution to the Schrödinger and Poisson Equations and on the Local Mobility Model

Tomasz Janik and Bogdan Majkusiak, *Member, IEEE*

***Abstract*— The effects of carrier energy quantization in the semiconductor surface region on performance of the metal-oxide-semiconductor (MOS) transistor are theoretically considered by comparison of results of a self-consistent solution to the Schrödinger and Poisson equations and the results of the classical description. The gate voltage dependencies of the surface potential and inversion layer charge density are compared. Using the local mobility model the relations between the electron effective mobility and the electric effective field obtained from the both descriptions are for the first time compared. The accuracy of the commonly used triangular well approximation is examined. This approximation is used for calculation of the transistor current–voltage ($I-V$) characteristics. Simulations are performed for MOS transistors with ultrathin oxides and highly doped substrates, in accordance with the state of the art of today's VLSI/ULSI technology.**

***Index Terms*—Modeling, MOSFET's, numerical analysis, quantization, semiconductor device modeling.**

I. Introduction

IN ORDER to fulfill scaling rules in design of the metal-oxide-semiconductor very large scale integration circuits (MOS/VLSI) [1], the gate oxide thickness is decreased and the substrate dopant concentration is increased. It has been shown theoretically [2] that the direct tunneling current through the gate oxide can be much less than the drain current even if the gate oxide thickness is reduced below 2 nm, provided the transistor channel is very short. Recently, the correct operation of an experimental MOS transistor with the gate oxide of 1.5 nm thickness has been presented [3]. Ultrathin oxides and high substrate doping result in an increase of the electric field at the semiconductor surface. A narrow and deep potential well is induced at the semiconductor surface in the strong inversion state. According to the quantum-mechanics theory, in such a potential well carriers create a two-dimensional electron gas (2-DEG), i.e., their kinetic energy in the direction perpendicular to the surface is quantizied, whereas the energy of their motion in the plain parallel to the semiconductor surface can take arbitrary values [4]–[6].

The 2-DEG is described by the system of Schrödinger and Poisson equations. It can be solved accurately by the use of numerical methods or analytically under some simplifying assumptions. Since the fundamental work of Stern [5], many authors have discussed the self-consistent solution to the above equation system for different physical states of the semiconductor region and in reference to different field effect devices [7]–[16]. These works revealed that the carrier concentration distribution resulting from the quantum-mechanics theory is qualitatively different than obtained from the classical theory based on the assumption of carrier energy continuum. We investigate the influence of this effect on the key relations for modeling the MOS transistor—the gate voltage dependencies of the surface potential and the electron charge density in the strong inversion state. At this point the accuracy of the triangular well approximation based on the assumption of a linear distribution of the potential [17] is also examined. Taking advantage of the conclusions from this investigation the triangular well approximation is used to calculate the current–voltage ($I-V$) characteristics of the MOS transistor. In order to explain the differences between $I-V$ characteristics obtained with the use of the "quantum" and "classical" models, an influence of the electron energy quantization on the transistor threshold voltage [18]–[21] must have been also included.

Since the carrier energy quantization affects distributions of the local electric field and carrier concentration, it can also affects the electron mobility in the inversion layer. There is a number of works devoted to the analysis of the effective electron mobility in the 2-DEG [7], [22]–[26]. Some of them give deep insight into the physical nature of the scattering mechanisms of carriers occupying quantizied subbands, as for example the acoustic and optical phonon scattering and the surface roughness scattering [25], [26], but they require very complex approach to the problem. On the other hand, classical MOS transistor simulators based on numerical solution to the transport equations use the local mobility models. The well known example is the mobility model implemented in the MINIMOS simulator [27]. It describes the local mobility as a function of both the distance from the semiconductor surface and the local transverse electric field in the inversion layer. The use of the "quantum" description of these magnitudes can be the simplest way to model the effect of carrier energy quantization on the effective mobility of electrons in the 2-DEG at the semiconductor surface. Using the MINIMOS 5 local mobility model, we show that the electron energy quantization practically does not affect the relation between the effective mobility and the effective electric field in the channel.

Manuscript received November 30, 1996; revised July 22, 1997. The review of this paper was arranged by Editor D. A. Antoniadis. This work was supported by the KBN, Poland, under Grants 8S50101107 and 8T11B04411.

The authors are with the Institute of Microelectronics and Optoelectronics, Warsaw University of Technology, Warsaw 00-662, Poland.

Publisher Item Identifier S 0018-9383(98)03667-3.

0018–9383/98$10.00 © 1998 IEEE

Authorized licensed use limited to: Chan Tan. Downloaded on September 01,2026 at 21:09:49 UTC from IEEE Xplore. Restrictions apply.

To make our analysis reliable, transistors with dopant concentrations and oxide thicknesses typical for aggressive scaling of today's MOS transistors are considered.

II. Physical Fundamentals

The gate-SiO_2-Si structure with the uniformly doped and $\langle 100 \rangle$ oriented p-type silicon is considered here, and calculations are done for the temperature $T = 300$ K. Most of the presented results concern the dopant concentration $N_A = 10^{17}$ cm^{-3} and the gate work function $\phi_{\rm mi} = 3.2$ eV (in reference to the SiO_2 conduction band), which can be interpreted as the aluminum gate or the poly-Si(n^+) gate.

Under the assumption that planar dimensions of the MOS transistor are much greater than the depth of the semiconductor potential well, the electrostatic characteristics of 2-DEG are described by the system of one-dimensional Schrödinger and Poisson equations

$$\frac{d^2\psi_{\rm ij}(x)}{dx^2} + \frac{2m_{\rm xj}}{\hbar^2}[E_{\rm ij} - V(x)]\psi_{\rm ij}(x) = 0 \tag{1}$$

$$\frac{d^2V}{dx^2} = \frac{q^2}{\varepsilon_s}[N_A^-(x) + n(x) - p(x)] \tag{2}$$

where $V(x)$ denotes the electrostatic potential energy, $\psi_{\rm ij}(x)$ is the normalized wave function representing an electron in the ijth state corresponding to the discrete energy level $E_{\rm ij}$, $m_{\rm xj}$ is the effective mass describing dynamics of the electron motion in the direction x perpendicular to the surface, N_A^- is the ionized acceptor concentration, and $n(x)$ and $p(x)$ are local values of the electron and hole concentrations, respectively.

Since the semiconductor band bending in the semiconductor surface region in the strong inversion state creates a potential well for electrons, the electron concentration distribution $n(x)$ must be calculated from the following:

$$n(x) = \sum_{j=1}^{2}\sum_{i=1}^{\infty} N_{\rm ij}|\psi_{\rm ij}(x)|^2 \tag{3}$$

$$N_{\rm ij} = \frac{n_{\rm vj}m_{\rm dj}k_BT}{\pi\hbar^2}\ln\left[1 + \exp\left(\frac{E_{\rm Fn} - E_{\rm ij}}{k_BT}\right)\right] \tag{4}$$

where (4) expresses the number of electrons occupying the ijth electron energy level [5], $m_{\rm dj}$ is the electron density-of-states mass, $n_{\rm vj}$ is the valley degeneracy, ε_s is the silicon electrical permittivity, and $E_{\rm Fn}$ is the electron quasi-Fermi level. Holes at the semiconductor surface regions are not confined by any potential well, then their energy is not quantizied and $p(x)$ can be expressed in the classical way.

The additional index j at $E_{\rm ij}$, $\psi_{\rm ij}$, $m_{\rm xj}$, $m_{\rm dj}$, $n_{\rm vj}$, and $N_{\rm ij}$ is used to denote the silicon conduction band structure. The constant energy surface for the silicon conduction band is composed of six ellipsoids located symmetrically on the k_x, k_y, and k_z axes in case of the $\langle 100 \rangle$ surface orientation. Therefore, the Schrödinger equation (1) must be solved separately for electrons from two ellipsoids ($j = 1$, $n_{v1} = 2$) with $m_{x1} = m_l$ (m_l is the longitudinal electron mass) and for electrons from four ellipsoids ($j = 2$, $n_{v2} = 4$) with $m_{x2} = m_t$ (m_t is the transverse electron mass). Consequently, two families of energy levels E_{i1} and E_{i2} exist, with $m_{d1} = m_t$ and $m_{d2} = (m_l m_t)^{1/2}$, respectively.

Since the self-consistent solution to (1) and (2) is very complex and computationally expensive, some simplified methods of analysis of the 2-DEG are applied. The most commonly used method is the triangular well approximation based on the assumption of a linear distribution of the potential in the semiconductor surface layer, i.e., the constant electric field in the inversion layer [17]. This assumption allows to decouple (1) and (2). As a result the analytical expression for the electron energy states is obtained

$$E_{\rm ij} = \left(\frac{\hbar^2}{2m_{\rm xj}}\right)^{1/3}\left[\frac{3}{2}\pi q F_{\rm eff}\left(i - \frac{1}{4}\right)\right]^{2/3} \qquad i = 1, 2, 3, \cdots \tag{5}$$

where $F_{\rm eff}$ denotes the effective electric field in the semiconductor surface region

$$F_{\rm eff} = -\frac{0.5Q_{\rm inv} + Q_{\rm depl}}{\varepsilon_s}. \tag{6}$$

Here, $Q_{\rm depl} = -[2\varepsilon_s Na(q\phi_s - k_BT)]^{1/2}$ is the depletion layer charge (ϕ_s denotes the surface potential) and $Q_{\rm inv}$ is the total inversion layer charge density

$$Q_{\rm inv} = -q\sum_{\rm ij} N_{\rm ij}. \tag{7}$$

In the case of classical description, where the quantization of electron energy is neglected, we use the Seiwatz–Green description of the semiconductor region [28], based on Fermi–Dirac statistics. According to it the electron concentration $n(x)$ is expressed in the following way:

$$n(x) = N_C F_{1/2}\left[u(x) - \frac{E_C - E_{\rm Fn}}{k_BT}\right] \tag{8}$$

where $F_{1/2}$ is the Fermi integral of the order 1/2 and $u = q\phi/k_BT$ is the normalized potential in the semiconductor. The relation between the normalized potential u and the position x is obtained from the following:

$$x(u) = \frac{k_BT\varepsilon_s}{2q^2n_iL_i}\int_0^u \frac{du}{F(u)} \tag{9}$$

where L_i is the Debye's length for the intrinsic semiconductor and $F(u)$ is given by [28]

$$F(u) = \left\{\frac{N_A}{n_i}\ln\frac{4 + \exp\left(u + \dfrac{E_F - E_A}{k_BT}\right)}{4 + \exp\left(\dfrac{E_F - E_A}{k_BT}\right)} + \frac{N_C}{n_i}\left[F_{3/2}\left(u - \frac{E_C - E_{\rm Fn}}{k_BT}\right) - F_{3/2}\left(-\frac{E_C - E_{\rm Fn}}{k_BT}\right)\right] + \frac{N_V}{n_i}\left[F_{3/2}\left(-u - \frac{E_F - E_V}{k_BT}\right) - F_{3/2}\left(-\frac{E_F - E_V}{k_BT}\right)\right]\right\}^{1/2}. \tag{10}$$

Authorized licensed use limited to: Chan Tan. Downloaded on September 01,2026 at 21:09:49 UTC from IEEE Xplore. Restrictions apply.

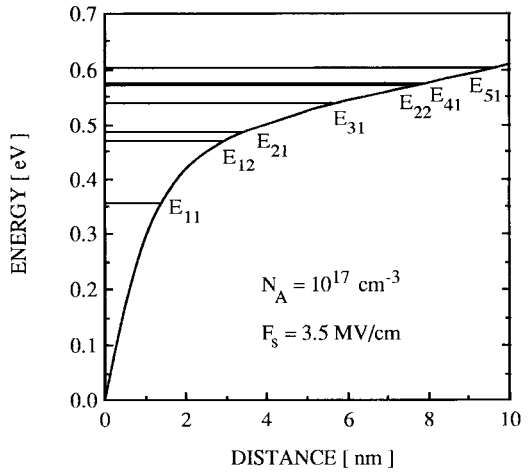


Fig. 1. The distribution of electron energy levels in the semiconductor surface region in the strong inversion state.

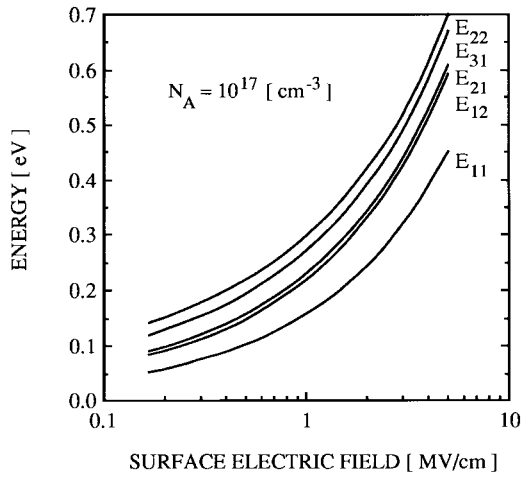


Fig. 2. The dependence of the electron energy levels at the silicon surface on the surface electric field.

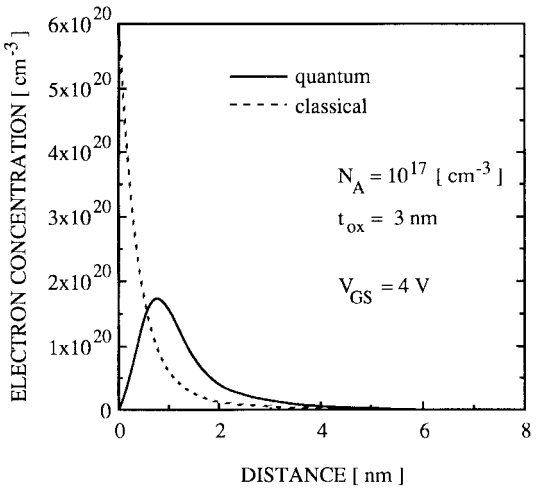


Fig. 3. The spatial electron concentration distribution according to the classical and the quantum models.

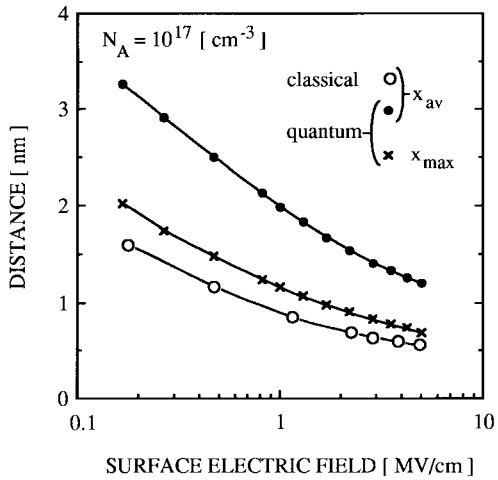


Fig. 4. The surface electric field dependence of the average electron distance from the semiconductor surface according to the different models and the position of the electron concentration maximum for the quantum model.

Here, N_C and N_V are the effective density of states in the semiconductor conduction band and valence band, respectively, n_i is the intrinsic concentration, E_C, E_V, and E_F denote the bottom of the conduction band, the top of the valence band and the energy of the Fermi level, respectively, and $F_{3/2}$ is the Fermi integral of the order 3/2.

The system of material constants used in our modeling is based on the work by Vankemmel *et al.* [29] and includes the expression for the shift of the majority carrier band edge proposed by Kuźmicz [30]. This system gives the following parameters for $N_A = 10^{17}$ cm^{-3} and $T = 300$ K: $E_g = 1.122$ eV, $n_i = 1.202 \times 10^{10}$ cm^{-3}, $m_l = 0.9163 m_0$ (m_0 is the free electron mass), and $m_t = 0.2245 m_0$.

III. Inversion Layer

Fig. 1 shows the obtained distribution of electron energy levels for a very large value of the electric field 3.5 MV/cm, induced at the semiconductor surface by the gate voltage 3.65 V in the case of the 3-nm oxide thickness. The case corresponds to the deep strong inversion state. As can be seen, the first energy level lies about 350 meV above the bottom of the conduction band and the distance between the first energy levels is of the order of 120 meV. In the case of highly doped substrate considered here, the carrier energy quantization is evident even at the beginning of the strong inversion state. The dependence of the energy levels E_{ij} on the surface electric field for $N_A = 10^{17}$ cm^{-3} is shown in Fig. 2. For instance, for the gate voltage 0.1 V inducing the surface field 0.17 MV/cm the first energy level lies about 50 meV above the bottom of the conduction band at the semiconductor surface and the distance between the first two energy levels is equal to about 30 meV. This means that the energy quantization should result in an increase of the threshold voltage of MOS transistors with highly doped semiconductor substrates [18]–[21].

Fig. 3 shows the electron concentration distributions resulting from the classical and quantum descriptions for the gate voltage $V_{GS} = 4$ V. The fundamental difference between them lies in the fact that a maximum of the electron concentration distribution is localized at the semiconductor surface according the classical model whereas it is shifted from the surface according the quantum model. Fig. 4 shows the position of the

Authorized licensed use limited to: Chan Tan. Downloaded on September 01,2026 at 21:09:49 UTC from IEEE Xplore. Restrictions apply.

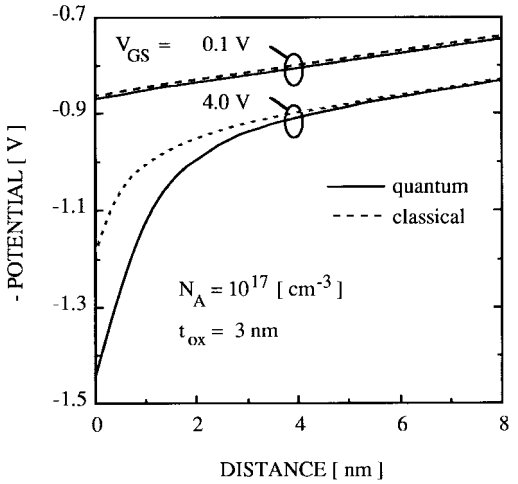


Fig. 5. The spatial distribution of the potential for different gate voltages and according to the different models.

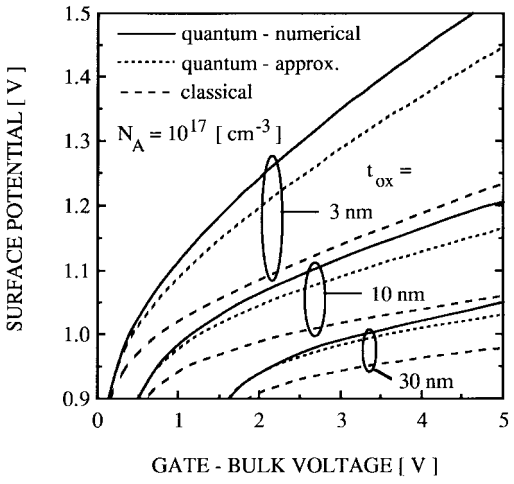


Fig. 6. The gate voltage dependence of the surface potential according to the classical, quantum-self consistent, and quantum-triangular well approximation models.

concentration peak $x_{\max}$ as a function of the surface electric field and compares the distribution centroids—the average distances of electrons from the semiconductor surface, obtained with the use of the classical and quantum descriptions. As can be seen, the average distance of electrons from the semiconductor surface resulting from the classical description is approximately half of that for the quantum description. This effect should results in overestimated semiconductor capacitance. On the other hand, an underestimated value of the average electron distance from the semiconductor surface can results in an underestimated value of the effective mobility. This problem will be considered in the next section.

Fig. 5 presents spatial distributions of the potential in the semiconductor for the gate voltages corresponding to the beginning of the strong inversion and the deep strong inversion state, respectively. The potential axis is reversed to illustrate the shape of the semiconductor conduction band bottom. As it is shown, the quantization of the electron energy results in a greater band bending at a given gate voltage in comparison to the classical case.

The differences between charge distributions obtained from the quantum and classical descriptions result in differences in the gate voltage dependencies of the surface potential and inversion layer charge, which are presented in Figs. 6 and 7, respectively. Curves marked as "quantum-numerical" denote results obtained with the use of the self-consistent procedure based on the quantum approach, whereas curves marked as "classical" denote results obtained by the use of the classical description. It can be seen, that the use of the classical description is a source of significant errors in the surface potential determination. For ultrathin oxides this error can be larger than 200 mV. It is worth noticing here, that an error in the calculation of the surface potential versus gate voltage results in an error in the calculation of the voltage drop on the oxide. This is especially critical for considerations concerning reliability of ultrathin oxides and other diagnostics issues. The relative error in the electron charge density resulted from the use of the classical description is about 7% at the gate-source voltage of 3 V and oxide thickness of 3 nm.

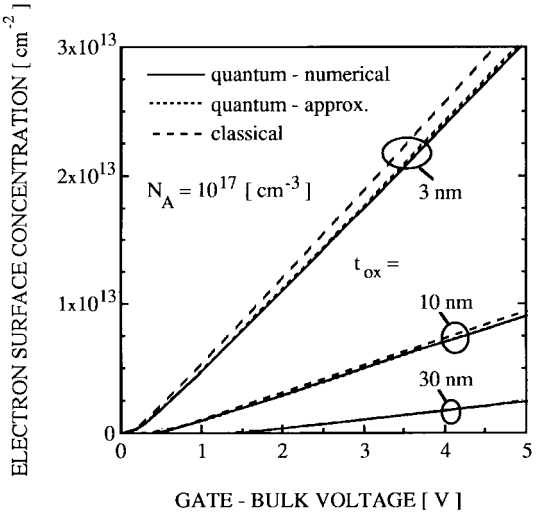


Fig. 7. The gate voltage dependence of the inversion layer charge density according to the different models.

Figs. 6 and 7 include also the results obtained by the use of the triangular well approximation. They are marked as the curves "quantum-approx". It can be concluded that this approximation is a source of significant errors in the surface potential versus the gate voltage calculations. However, the gate voltage dependence of the inversion layer charge density calculated by the use of the triangular well approximation differs slightly from the exact numerical results. Therefore, one can conclude that the triangular well approximation is accurate enough for calculating the gate voltage dependence of the inversion layer charge.

IV. Electron Mobility

Since the electron energy quantization results in changed distributions of potential and transverse electric field, it is interesting to see how this phenomenon influences the electron mobility. The direct effect of the energy quantization on the electron mobility can be investigated by consideration of the rates of different scattering mechanisms, as studied in [24],

Authorized licensed use limited to: Chan Tan. Downloaded on September 01,2026 at 21:09:49 UTC from IEEE Xplore. Restrictions apply.

[25] in reference to the phonon scattering. In this paper we use a simpler way—we will see how the changed description of the semiconductor region influences the distribution of the local mobility and the average mobility μ_{eff}. For this purpose we apply the local mobility model used in the MINIMOS 5 simulator, which generates the effective mobility versus the effective transverse electric field relation in agreement with experimentally measured universal dependence [27].

The effective mobility represents the local $\mu_n(x)$ mobility averaged over the total population of carriers $n(x)$

$$\mu_{\text{eff}} = \frac{\int \mu_n(x) n(x)\, dx}{\int n(x)\, dx} \tag{11}$$

while the effective electric field E_{eff} represents the local transverse electric field $E_x(x)$ averaged over the electron population

$$E_{\text{eff}} = \frac{\int E_x(x) n(x)\, dx}{\int n(x)\, dx}. \tag{12}$$

Fig. 8 presents the spatial distributions of local electron mobility in the semiconductor for two different gate voltages. In the case of weak inversion ($V_{\text{GB}} = 0.1$ V), there is almost no difference in the band bending as well as in the local electric field in the semiconductor between the quantum and classical description (see Fig. 5) and this is why there are also no differences in the local mobility distributions. The electric field is too small to influence the value of mobility. An increase of the local mobility for distances greater than 10 nm results from the characteristics of the MINIMOS 5 local mobility model [27] and can be interpreted as result of a decrease of the electron scattering on defects generated by the surface and distributed interior the semiconductor. 10 nm is a value which can be interpreted as the lattice disturbance depth. In the case of a very strong inversion ($V_{\text{GB}} = 4$ V), for which the electric field below 5 nm from the surface is high enough to affect the electron mobility, there is also a significant difference in the local mobility distributions.

Fig. 9 shows the effective field dependence of the effective electron mobility calculated on the base of the definitions (11) and (12). The surprising fact is that in spite of the significant changes in spatial distributions of the electron concentration, electric field and the local mobility due to the energy quantization, the relation between the average mobility and average electric field remains practically unchanged. The only difference relies on the shift of points corresponding to the same values of the surface potential.

V. Electrical Characteristics

The integral of the product of electron concentration and the local electron mobility over the space charge region determines the normalized output conductance of the MOS transistor at very small drain-source voltages

$$g_{d0}\frac{L}{W} = q\int_0^{x_{\text{inv}}} \mu_n(x) n(x)\, dx = -\mu_{\text{eff}} Q_{\text{inv}}. \tag{13}$$

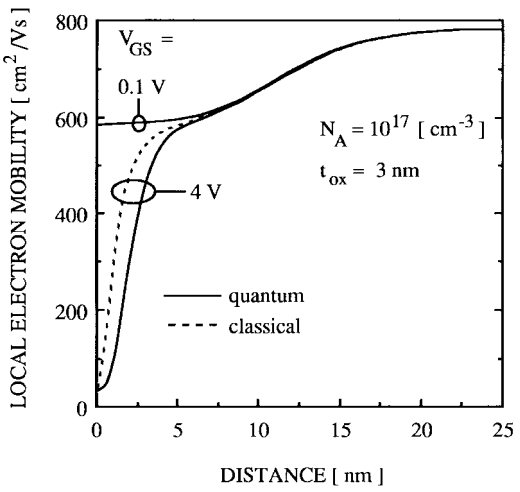


Fig. 8. The spatial distribution of the local electron mobility in the semiconductor according to the classical and quantum descriptions of the semiconductor region.

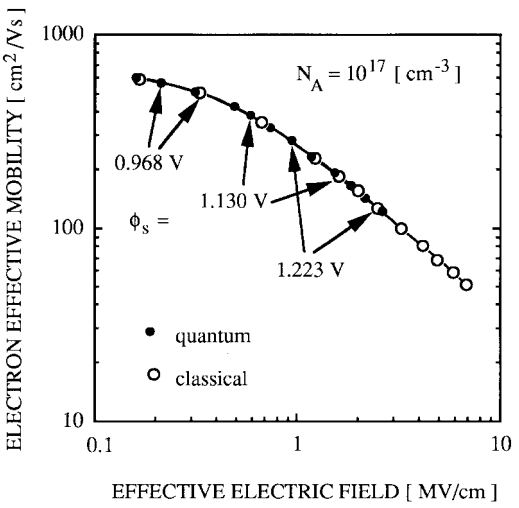


Fig. 9. The dependence of effective electron mobility on the effective electric field for the classical and quantum descriptions of the semiconductor region.

Fig. 10 shows spatial distributions of the product of local values of the electron mobility and electron concentration according to the classical and quantum descriptions. For the deep strong inversion state the degradation of the local mobility near the surface (as shown in Fig. 8) results in a maximum in this product in some distance from the surface even for the classical case. In spite of significant differences in spatial distributions of the product $\mu(x)n(x)$ for the quantum and classical models, integration (13) gives only a small shift on the voltage axis in the output conductance g_{d0} versus the gate-bulk voltage dependence, as shown in Fig. 11. This shift resulting from the energy quantization can be interpreted as the increase of the threshold voltage, which is usually extracted from the measured $g_{d0}(V_{\text{GS}})$ dependence.

Simulation of the drain current of the MOS transistor at a nonzero V_{DS} voltage, based on self-consistent solution of the Schrödinger and Poisson equations in direction x and along the

Authorized licensed use limited to: Chan Tan. Downloaded on September 01,2026 at 21:09:49 UTC from IEEE Xplore. Restrictions apply.

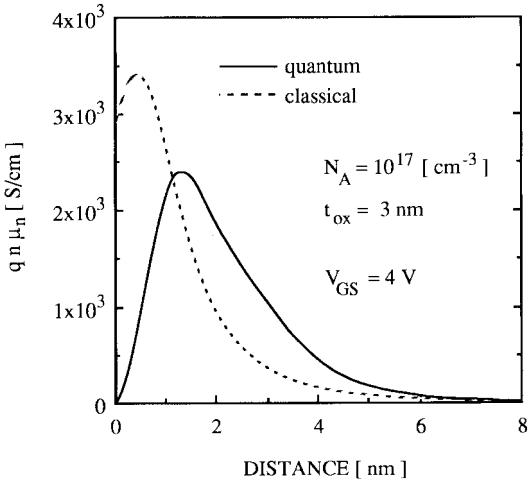


Fig. 10. The spatial distributions of the product of the electron concentration and local electron mobility in the semiconductor.

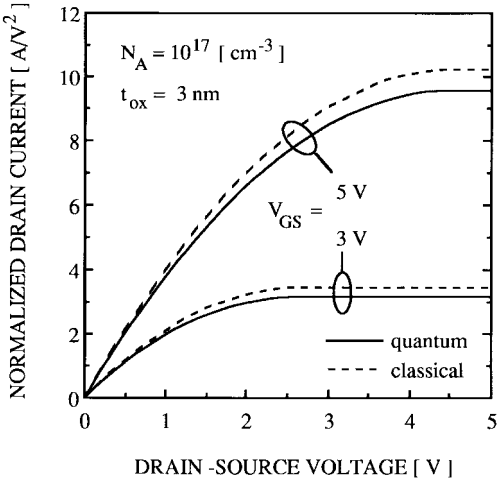


Fig. 12. The output characteristics of an MOS transistor according to the different models.

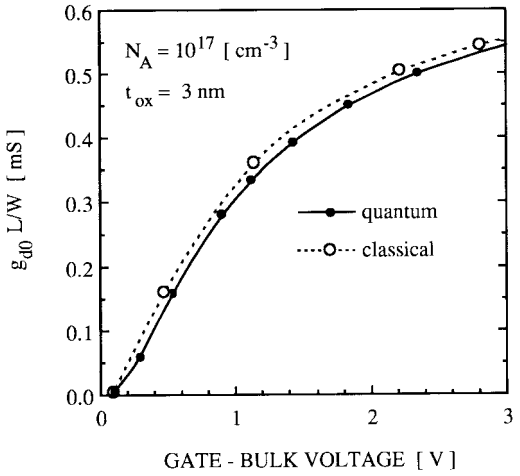


Fig. 11. The normalized output conductance of the MOS transistor versus the gate voltage according to the different models.

channel is computationally very expensive, whereas as shown in the previous section the triangular well approximation is accurate enough to model the gate voltage dependence of the inversion layer charge. Therefore this approximation can be also used for modeling the gate voltage dependence of the drain current.

In order to show the effect of energy quantization on the drain current, we use the Pao–Sah model [31]. According to this model the drain current is obtained as the integral of the inversion layer charge over the quasi-Fermi level split from the source end to the drain end of the channel

$$I_D = -\frac{W}{L}\frac{k_BT}{q}\int_{v_{SB}}^{v_{SB}+v_{DS}} \mu_{\text{eff}}(y)Q_{\text{inv}}(y)\,du_n \qquad (14)$$

where W and L are the gate width and length, respectively, y is the direction from the source to the drain, $v_{SB} = qV_{SB}/k_BT$ and $v_{DS} = qV_{DS}/k_BT$ are the normalized source-bulk and drain-source voltages, respectively.

In the quantum model the inversion charge density is calculated from the expression (7). According to the classical model the inversion layer charge Q_{inv} at a given plane y of the channel is obtained by integrating the electron charge density over the space charge region from the beginning of the inversion region x_{inv}, where $u(x_{\text{inv}}) = u_F = q\phi_F/k_BT$ (u_F is the normalized Fermi potential), to the surface, where $u = u_s = q\phi_s/k_BT$ (u_s is the normalized surface potential)

$$\begin{aligned} Q_{\text{inv}}(y) &= -q\int_0^{x_{\text{inv}}(y)} n(x,y)\,dx \\ &= -qL_iN_C\int_{u_F}^{u_S}\frac{F_{1/2}\left(u-u_n-\dfrac{E_C-E_F}{k_BT}\right)}{F(u,u_n)}\,du. \end{aligned} \qquad (15)$$

Here, $u_n = (E_F - E_{\text{Fn}})/k_BT$ is the split of the quasi-Fermi levels in the plane y.

In order to investigate the effect of the use of the classical description of the semiconductor region on the drain current, we assume that μ_{eff} is constant along the channel. We will show the drain current normalized by $\mu_{\text{eff}}C_{\text{ox}}W/L$, where C_{ox} is the gate oxide capacitance. Fig. 12 presents the calculated output characteristics of the MOS transistor with the ultrathin 3 nm gate oxide. It is shown that for given drain-source and gate-source voltages the carrier energy quantization reduces the drain current. This decrease results from the reduction of the inversion layer charge caused by the carrier energy quantization (Fig. 7).

The relative error of calculation of the drain current in saturation, caused by the use of the classical description is shown in Fig. 13 for different substrate dopant concentrations and for two cases: at $V_{GS} = 3$ V and at $V_{GS} - V_T = 2$ V, where V_T is the threshold voltage. The error depends strongly on the oxide thickness and the substrate doping. In case of low doping levels the error significantly increases as the oxide thickness is reduced. It results from the increase of the surface electric field which causes the stronger split of the electron energy levels (Fig. 2). In the case of high doping levels the error calculated for the case of the constant gate voltage

Authorized licensed use limited to: Chan Tan. Downloaded on September 01,2026 at 21:09:49 UTC from IEEE Xplore. Restrictions apply.

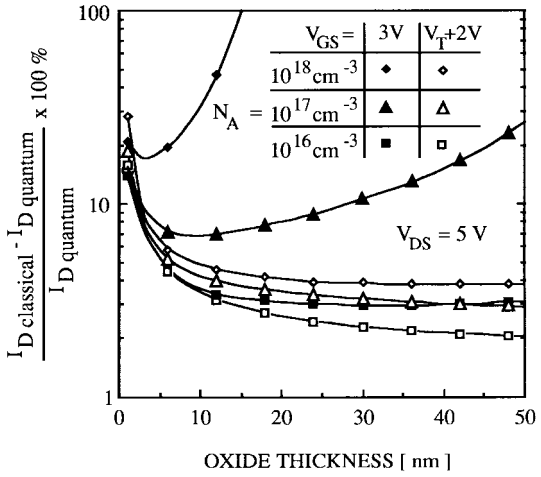


Fig. 13. The relative error of calculation of the drain current in saturation caused by the use of the classical description.

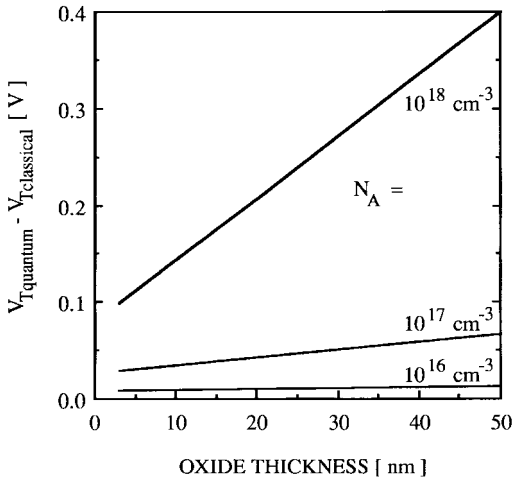


Fig. 14. The difference between threshold voltage resulting from the quantum and classical descriptions of the semiconductor surface region.

increases also for thick oxides which may have practical implication as the drain current overestimation in the parasitic channel in the highly doped semiconductor region under the field-oxide isolation. This unexpected effect is a consequence of the increase of threshold voltage caused by the carrier energy quantization [18]–[21]. To compare the error in the drain current simulation at the constant $V_{\rm GS} - V_T$ voltage we calculated the threshold voltage independently for the classical and quantum descriptions and for different dopant concentrations and oxide thicknesses. The definition of the threshold voltage used in modeling as well as the method of calculations is given in Appendix. To obtain the threshold voltage for the quantum description we developed the accurate model derived in [19] to a new formula for V_T (A3). The shift in the threshold voltage due to the energy quantization increases with increasing the oxide thickness and the dopant concentration, as shown in Fig. 14. As can be seen in Fig. 13, the error in drain current calculation at the constant $V_{\rm GS} - V_T$ difference, resulting from the use of the classical description, is equal to about $2 \div 4\%$ for thick oxides and reaches several percents for ultrathin oxides, in dependence on the dopant concentration. Therefore, one can conclude that the drain current model of the MOS transistor with ultrathin oxide should take into account not only the threshold voltage shift, which is significant for high doping levels, but also the change of the inversion layer charge as a result of the electron energy quantization.

VI. Conclusions

In the aggressively scaled MOS transistors with ultrathin gate oxides and highly doped semiconductor substrates the quantization of carrier energy becomes an important effect from the onset of the strong inversion state. First of all, it affects strongly the electron distribution in the semiconductor surface region. Carriers are shifted from the surface which is reflected in the increase of the inversion layer width. The total inversion charge induced at a given gate voltage is decreased, whereas the band bending at the semiconductor surface layer is increased. Therefore, the carrier energy quantization especially influences the gate voltage dependence of the surface potential, especially in the case of ultrathin gate oxides, for which the error in determination of the surface potential can reach 200 mV at the gate voltage of the order of a few volts. This leads also to an overestimation of the oxide voltage by the same error.

The resulting difference in the band bending and the local electric field in the semiconductor changes also the local mobility distribution. However, one of the most essential result obtained in this work is that the classical and the quantum descriptions of the semiconductor region do not influence the dependence of the electron effective mobility on the effective transverse electric field in the channel. The only difference relies on the shift of the points corresponding to the same values of the surface potential.

One of the well known consequences of the energy quantization in the inversion layer is the increase of the threshold voltage, which has already been considered in the literature as a consequence of high doping of the substrate. We showed this effect as a shift in the calculated gate voltage dependence of the output conductance as well as by modeling its dependence on the oxide thickness and dopant concentration. For this purpose we developed the accurate threshold voltage model [19] to a new equation for the threshold surface potential.

Although the consideration of the threshold voltage is a well known effect, we needed it to analyze the error in the drain current modeling. We conclude that the energy quantization effect on the threshold voltage for highly doped semiconductors is especially important for thick oxides. The increased threshold surface potential results in the increased depletion layer charge density, which is multiplied by the oxide thickness in the formula for the threshold voltage. Therefore, the neglection of the energy quantization should lead to the underestimation of the threshold voltage of the parasitic channel in the highly doped semiconductor under the field-oxide isolation. If the oxide thickness is of the order of few nanometers, the threshold voltage shift is smaller, but even a small increase of the gate voltage above the threshold voltage drops mainly on the semiconductor and induces high electric

Authorized licensed use limited to: Chan Tan. Downloaded on September 01,2026 at 21:09:49 UTC from IEEE Xplore. Restrictions apply.

field and a deep potential well at its surface. The carrier energy quantization in such a well leads to the errors in drain current modeling of the order of several percents, in dependence on the oxide thickness and substrate doping.

An accurate analysis of the carrier energy quantization requires numerical solution to the system of Schrödinger and Poisson equations. As the presented analysis shows, such a rigorous approach is necessary for the surface potential calculations in the case of very thin gate oxides. However, the gate voltage dependence of the inversion layer charge may be modeled quite accurately by the use of the triangular well approximation. Consequently, this approximation may be both an accurate and efficient method for modeling the I–V characteristics of the MOS transistor.

APPENDIX

In [19] we applied the definition of the threshold voltage of the MOS transistor in the form which can be used both for the classical and the quantum considerations

$$\left|\frac{dQ_{\rm inv}}{d\phi_s}\right|_{\phi_s=\phi_{sT}} = \left|\frac{dQ_{\rm depl}}{d\phi_s}\right|_{\phi_s=\phi_{sT}} \tag{A1}$$

where ϕ_{sT} is the surface potential at the threshold state, and the depletion layer charge is expressed as

$$Q_{\rm depl} = -\sqrt{\left[\varepsilon_s\frac{k_BT}{qL_i}F(u_F)\right]^2 + 2\varepsilon_s qN_A(\phi_s-\phi_F)}. \tag{A2}$$

Then, the threshold voltage is obtained from

$$V_T = V_{FB} + \phi_{sT} - \frac{Q_{\rm depl}(\phi_{sT}) + Q_{\rm inv}(\phi_{sT})}{\varepsilon_{\rm ox}}t_{\rm ox}. \tag{A3}$$

To obtain the inversion layer charge density given by (7) and (4) according to the quantum description, the electron energy levels $E_{\rm ij}$ in the potential well induced at the semiconductor surface in the depletion state must be found. Approximating the shape of the well by a truncated parabola and using WKB approximation [32] we derived in [19] the following expression for electron energy levels

$$x_d\sqrt{\frac{m_{\rm xj}q\phi_s'}{2\hbar^2}}\left[\sqrt{\frac{E_{\rm ij}}{q\phi_s'}} + \left(1-\frac{E_{\rm ij}}{q\phi_s'}\right)\ln\left(\frac{1-\sqrt{\dfrac{E_{\rm ij}}{q\phi_s'}}}{\sqrt{1-\dfrac{E_{\rm ij}}{q\phi_s'}}}\right)\right] = \left(i-\frac{1}{4}\right)\pi \tag{A4}$$

where x_d is the depletion later width and ϕ_s' is the parameter of this parabola and is equal $\phi_s - k_BT/q$ for low doping levels. In this work we develop the condition for the threshold state (A1) by analytical differentiating (A2) and (A4) and finally we obtain the equation for the threshold surface potential ϕ_{sT} in the form

$$\frac{\varepsilon_s N_A}{\sqrt{\left[\varepsilon_s\dfrac{k_BT}{qL_i}F(u_F)\right]^2 + 2\varepsilon_s qN_A(\phi_s-\phi_F)}} = \sum_{\rm ij}\frac{qm_{\rm vj}m_{\rm dj}}{\pi\hbar^2\left[1+\exp\left(-\dfrac{E_{\rm Fn}-E_{\rm ij}}{k_BT}\right)\right]} \cdot \left[\frac{\left[\sqrt{e_{\rm ij}}+(1-e_{\rm ij})\ln\left(\dfrac{1-\sqrt{e_{\rm ij}}}{\sqrt{1-e_{\rm ij}}}\right)\right]^3}{\left(i-\frac{1}{4}\right)\pi\hbar\sqrt{\dfrac{N_A}{\varepsilon_s m_{\rm xj}}}\ln\left(\dfrac{1-\sqrt{e_{\rm ij}}}{\sqrt{1-e_{\rm ij}}}\right)} - (1-e_{\rm ij})\right] \tag{A5}$$

where $e_{\rm ij} = E_{\rm ij}/q\phi_s'$.

The condition (A1) applied for the classical description leads to the equation

$$N_C F_{1/2}\left[\frac{q\phi_{sT}-(E_C-E_{\rm Fn})}{k_BT}\right] = N_A \tag{A6}$$

which can be transformed to the well known form according to the Maxwell–Boltzmann statistics if the Fermi integral is changed by the exponential function.

REFERENCES

[1] R. H. Dennard, F. H. Gaensslen, H. N. Yu, V. L. Rideout, E. Bassous, and A. Le Blanc, "Design of ion-implanted MOSFET's with very small physical dimensions," *IEEE J. Solid-State Circuits*, vol. SSC-9, pp. 256–268, Sept. 1974.

[2] B. Majkusiak, "Gate tunnel current in an MOS transistor," *IEEE Trans. Electron Devices*, vol. 37, pp. 1087–1092, Apr. 1990.

[3] H. Sasaki, M. Ono, T. Yoshitomi, T. Ohguro, S. Nakamura, M. Saito, and H. Iwai, "1.5 nm direct-tunneling gate oxide Si MOSFET's," *IEEE Trans. Electron Devices*, vol. 43, pp. 1233–1241, Aug. 1996.

[4] F. Stern and W. E. Howard, "Properties of semiconductor surface inversion layers in the electric quantum limit," *Phys. Rev.*, vol. 163, pp. 816–835, 1967.

[5] F. Stern, "Self-consistent results for n-type Si inversion layers," *Phys. Rev. B.*, vol. 5, pp. 4891–4899, 1972.

[6] A. P. Gnädinger and H. E. Talley, "Quantum mechanical calculation of the carrier distribution and the thickness of the inversion layer of a MOS field-effect transistor," *Solid State Electron.*, vol. 13, pp. 1301–1309, 1970.

[7] F. Stern, "Quantum properties of surface space-charge layers," *CRC Crit. Rev. Solid-State Sci.*, vol. 4, pp. 499–514, 1974.

[8] C. Moglestue, "Self-consistent calculation of electron and hole inversion charges at silicon-silicon dioxide interfaces," *J. Appl. Phys.*, vol. 59, pp. 3175–3183, 1986.

[9] Y. Ohkura, "Quantum effects in Si n-MOS inversion layer at high substrate concentration," *Solid State Electron.*, vol. 33, pp. 1581–1585, 1990.

[10] S. Imanaga and Y. Hayafuji, "Monte Carlo modeling of electron transport in a Si metal-oxide-semiconductor inversion layer including quantum state and bulk scattering," *J. Appl. Phys.*, vol. 70, pp. 1522–1530, 1991.

[11] Y. Ando and T. Itoh, "Analysis of charge control in pseudomorphic two-dimensional electron gas field-effect transistors," *IEEE Trans. Electron Devices*, vol. 35, pp. 2295–2301, Dec. 1988.

[12] M. B. Patil and H. Morko, "Self-consistent calculation of electron density in a two-channel modulation-doped structure," *Solid State Electron.*, vol. 33, pp. 99–104, 1990.

[13] A. A. Frantsuzov, S. A. Okhonin, and A. P. Pogosov, "The influence of quantization on the space-charge layer capacitance of Si in strong accumulation," *Phys. Stat. Sol. B*, vol. 136, pp. 241–249, 1986.

[14] J. Suñé, P. Olivo, and B. Riccó, "Self-consistent solution of the Poisson and Schrödinger equations in accumulated semiconductor-insulator interfaces," *J. Appl. Phys.*, vol. 70, pp. 337–345, 1991.

Authorized licensed use limited to: Chan Tan. Downloaded on September 01,2026 at 21:09:49 UTC from IEEE Xplore. Restrictions apply.

[15] J. A. López-Villanueva, I. Melchor, F. Gàmiz, J. Banqueri, and J. A. Jimenez-Tejada, "A model for the quantized accumulation layer in metal-insulator-semiconductor structures," *Solid State Electron.*, vol. 38, pp. 203–210, 1995.
[16] C.-Y. Hu, S. Banerjee, K. Sadra, B. G. Streetman, and R. Sivan, "Quantization effects in inversion layers of PMOSFET's on Si (100) substrates," *IEEE Electron Device Lett.*, vol. 17, pp. 276–278, June 1996.
[17] D. Colman, R. T. Bate, and J. P. Mize, "Mobility anisotropy and piezoresistance in silicon p-type inversion layers," *J. Appl. Phys.*, vol. 39, pp. 1923–1931, 1968.
[18] M. J. Dort, P. H. Woerlee, A. J. Walker, C. A. H. Juffermans, and H. Lifka, "Influence of high substrate doping levels on the threshold voltage and the mobility of deep-submicrometer MOSFET's," *IEEE Trans. Electron Devices*, vol. 39, pp. 932–938, 1992.
[19] T. Janik and B. Majkusiak, "Influence of carrier energy quantization on threshold voltage of metal oxide semiconductor transistor," *J. Appl. Phys.*, vol. 75, pp. 5186–5190, 1994.
[20] G. Chindalore, S. A. Hareland, S. Jallepalli, A. F. Tasch, Jr., C. M. Maziar, V. K. F. Chia, and S. Smith, "Experimental determination of threshold voltage shifts due to quantum mechanical effects in MOS electron and hole inversion layers," *IEEE Electron Device Lett.*, vol. 18, pp. 206–208, May 1997.
[21] S. Jallepalli, J. Bude, W. K. Shih, M. R. Pinto, C. M. Maziar, and A. F. Tasch, Jr., "Effects on quantization on the electrical characteristics of deep submicron p- and n-MOSFET's," in *Symp. VLSI Technol. Dig., Tech. Papers*, 1996, pp. 138–139.
[22] K. Lee, J. S. Choi, S. P. Sim, and C. K. Kim, "Physical understanding of low-field carrier mobility in silicon MOSFET inversion layer," *IEEE Trans. Electron Devices*, vol. 38, pp. 1905–1911, Aug. 1991.
[23] T. Ando, A. B. Fowler, and F. Stern, "Electronic properties of two-dimensional systems," *Rev. Mod. Phys.*, vol. 54, pp. 437–672, 1982.
[24] P. J. Price, "Two-dimensional electron transport in semiconductor layers: Part I—Phonon scattering," *Ann. Phys.*, vol. 133, pp. 217–239, 1981.
[25] K. Masaki, C. Hamagushi, K. Taniguchi, and M. Iwase, "Electron mobility in Si inversion layers," *Jpn. J. Appl. Phys.*, vol. 28, pp. 1856–1863, 1989.
[26] S. Takagi, A. Toriumi, M. Iwase, and H. Tango, "On the universality of inversion layer mobility in Si MOSFET's: Part II—Effects of surface orientation," *IEEE Trans. Electron Devices*, vol. 41, pp. 2363–2368, Dec. 1994.
[27] S. Selberherr, W. Hänsch, M. Seavey, and J. Slotboom, "The evolution of the MINIMOS mobility model," *Solid State Electron.*, vol. 33, pp. 1425–1436, 1990, and the *MINIMOS 5-Manual*.
[28] R. Seiwatz and M. Green, "Space charge calculations for semiconductors," *J. Appl. Phys.*, vol. 29, p. 1034, 1958.
[29] R. Vankemmel, W. Schoenmaker, and K. De Meyer, "A unified wide temperature range model for the energy gap, the effective carrier mass and intrinsic concentration in silicon," *Solid State Electron.*, vol. 36, pp. 1379–1384, 1993.
[30] W. Kuźmicz, "Recent advances in understanding of the minority carrier transport in heavily doped silicon," in *Physics of Semiconductor Devices, Proceedings of II International Workshop.* Delhi, India: Tata-McGraw-Hill, 1983.
[31] H. C. Pao and C. T. Sah, "Effects of diffusion current on characteristics of metal-oxide (insulator)-semiconductor transistors," *Solid State Electron.*, vol. 9, pp. 927–937, 1966.
[32] E. Merzbacher, *Quantum Mechanics.* New York: Wiley, 1970.



Tomasz Janik was born in Warsaw, Poland, in 1966. He received the M.Sc. and Ph.D. degrees (with honors) in electrical engineering from Warsaw University of Technology, Poland, in 1991 and 1996, respectively.

He joined the Institute of Microelectronics and Optoelectronics, Warsaw University of Technology, in 1991, where he is currently an Assistant Professor. In 1996, he worked for six months as a Research Scientist at the National Microelectronics Research Center, Cork, Ireland. His research domain is the physics of semiconductor devices. Modeling of the MOS devices is the main area of his interest. His research has been especially focused on quantum effects in MOS transistors. Recently, he has been working on numerical analysis of electrostatic properties of MOS SOI transistors.

Dr. Janik was awarded a scholarship from the Foundation for Polish Science in 1996.



Bogdan Majkusiak (M'92) was born in Warsaw, Poland, in 1955. He received the M.Sc. degree (with honors) from the Faculty of Electronics, Warsaw University of Technology, Poland, in 1979, and the Ph.D. degree (with honors) and the D.Sc. degree from Warsaw University of Technology in 1985 and 1991, respectively.

He joined the Institute of Microelectronics and Optoelectronics, Warsaw University of Technology, in 1978, where he has been a Professor since May 1995. Currently, he is a V-Dean for Academic Affairs at the Faculty of Electronics and Information Technology. From 1993 to 1996, he was Head of the Microelectronics Specialty at the Faculty of Electronics and Information Technology. In 1984, he spent six months at the Institute of Electron Technology (ITE CEMI), Warsaw, and, in 1992, he worked for six months as a Visiting Research Scientist at Carnegie Mellon University, Pittsburgh, PA. His research interests include physics, modeling, characterization, and technology of the metal-insulator-semiconductor devices with emphasis on problems accompanying ultrathin insulators and quantum-mechanical phenomena. He has authored or coauthored more than 70 technical papers. He is also the author of several invited papers.

Dr. Majkusiak received the Award of Ministry of Science and Higher Education for outstanding achievements in research work in 1987.

Authorized licensed use limited to: Chan Tan. Downloaded on September 01,2026 at 21:09:49 UTC from IEEE Xplore. Restrictions apply.